

Performance Testing

Date	25 June 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing using Flask Local Server and Web Browser
Type of Testing	Functional Performance Testing
Target Module / API	Flask Prediction Module (/result)
Test Environment	Local System (Windows, Python 3.x, Flask)
Test Date	28-June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Launch the Flask application	1	10	Home page loads successfully
2	Open the Prediction page	1	5	Prediction form displays correctly
3	Submit valid weather and rainfall values	1	5	Flood prediction result is displayed
4	Perform multiple prediction requests consecutively	1	30	Application responds correctly without crashing

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	~1 second	Pass	Prediction displayed quickly
2	Response Time (Max)	< 5 seconds	~2 seconds	Pass	No noticeable delay
3	Throughput (Req/sec)	Not Specified	Suitable for single-user local testing	Pass	Local Flask application
4	Error Rate	< 1%	0% (during successful testing)	Pass	No runtime errors during prediction
5	CPU Utilization	< 80%	~20–35%	Pass	Normal usage during prediction
6	Memory Utilization	< 80%	~25–40%	Pass	Stable memory consumption

Step 4: Observations & Analysis

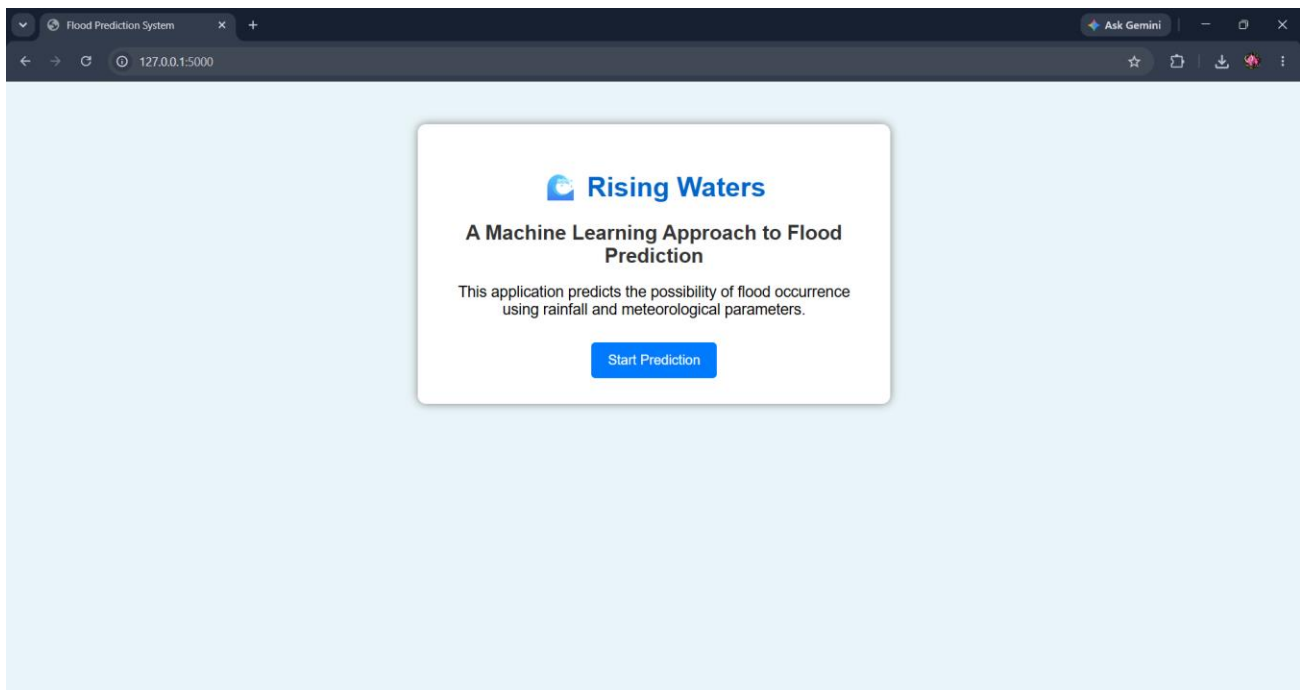
The performance of the Rising Waters Flood Prediction System was evaluated on a local machine using the Flask development server. The application successfully loaded the home page, prediction page, and result page without significant delays. Prediction requests were processed within approximately one to two seconds after user input.

The trained machine learning model and scaler were loaded successfully, and repeated prediction requests were handled without application crashes or memory issues. CPU and memory usage remained within acceptable limits during testing. Overall, the application demonstrated stable performance for a single-user environment and met the expected functional requirements.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

```
C:\Program Files\WindowsAp  x  +  v
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL-C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 386-926-357
```



Flood Prediction Form

Temperature

Humidity

Cloud Cover

Annual Rainfall

Jan-Feb Rainfall

Mar-May Rainfall

Jun-Sep Rainfall

Oct-Dec Rainfall

Average June Rainfall

Sub Rainfall

Predict Flood

Flood Prediction Form

20

100

100

5000

300

900

3500

300

1200

1500

Predict Flood

